

Abyud Somashekara Shetty

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PROFILE

Computer Science undergraduate drawn to backend and systems work: Spring Boot services over PostgreSQL, Python data and LLM pipelines, and real-time sensor software, with some 3D and graphics work alongside. Tries to get the fundamentals right along the way, through invariants in the database, automated tests, clear documentation, and debugging one hypothesis at a time. Learns by building, and enjoys work that leads into unfamiliar territory.

EDUCATION

Bachelor of Technology in Computer Science

Sep 2023 – Present

PES University

- **Coursework:** Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks, Cloud Computing, Machine Learning, Image Processing, Augmented and Virtual Reality, Digital Twins

EXPERIENCE

Summer Intern

Jun 2026 – Jul 2026

Centre for Augmented and Virtual Environments (CAVE) Lab, PES University 🏠

- Built real-time full-body motion capture from 10 MPU6050 IMUs on distributed ESP32 nodes, streaming orientation over ESP-NOW, UDP, and WebSocket into a Kotlin SlimeVR server that solves the skeleton with inverse kinematics.
- Retargeted tracker quaternions onto a Mixamo-rigged avatar in Three.js using bind poses and inverse parent-world rotations, and implemented foot-lock root motion for walking and sitting.
- Traced rotation jumps to quaternion sign flips and locked in the fixes with Kotlin regression tests; containerized the stack with Docker Compose for one-command setup.

PROJECTS

AEGIS - Financial Risk & Operations Control Plane

Aug 2026

- Engineered a financial transaction and risk-control platform with Java 21, Spring Boot 3.5, Spring Data JPA, and PostgreSQL, exposing REST APIs to a React/TypeScript control room.
- Prevented double-spends with pessimistic row locking and database-enforced idempotency, built a transactional outbox with bounded retries and scheduled recovery, and applied BFS/DFS analysis to transaction networks.
- Wrote 60 integration tests with JUnit 5 and MockMvc; containerized the stack with Docker Compose.

Buy or Wait? - LLM-Assisted Affordability Agent

Sep 2026

- Built a Python agent that rebuilds each user's financial timeline and forecasts 90-day balances to recommend full, partial, instalment, or wait decisions across 250 requests, using exact decimal arithmetic and dated FX rates.
- Used LLMs only to extract cited facts, rejecting any whose quote, amount, or date was not verbatim in the source.
- Gated output behind a verifier that replays every plan, backed by 103 tests including prompt-injection cases; the full run cost about \$0.011 per request and reproduces from cache with zero model calls.

Healthcare Records Management System

Dec 2025

- Developed a full-stack records platform with Django REST Framework, React/Redux, PostgreSQL, and Celery, exposing REST APIs for patient records, appointments, notifications, and reporting.
- Implemented JWT authentication, multi-factor login, role-based control, AES-256 record encryption, and immutable auditing.
- Automated CI/CD in GitHub Actions with unit, integration, system, and Locust load tests for 500+ concurrent users and 10K+ records, plus database performance benchmarks.

VRATA - Privacy-Preserving Reliability Estimation

Jul 2026

- Contributed to a research system that estimates fleet-wide Weibull reliability for wind turbines across mutually distrusting operators using malicious-secure multi-party computation (MP-SPDZ/MASCOT), so no operator ever shares raw failure data.
- Built the evaluation harness benchmarking the secure result against four baselines, including local-only estimation, method of moments, differential privacy, and federated averaging, and containerized multi-party orchestration with Docker.

MiniRAFT - Distributed Collaborative Drawing Board

Apr 2026

- Contributed to a three-node cluster running a from-scratch RAFT consensus implementation in Node.js and WebSockets, covering leader election, log replication, heartbeats, and term management behind a shared drawing board.
- Cluster handles leader discovery, request forwarding, and replica catch-up, re-electing automatically when a node is killed and rejoins, with health and replication endpoints exposed through Docker Compose.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, Kotlin, C++

Backend & APIs: Spring Boot, Spring Data JPA/Hibernate, Django REST Framework, FastAPI, Flask, Celery, REST APIs, WebSockets, JWT & RBAC

Databases: PostgreSQL, MySQL, MongoDB, Firebase Realtime Database, H2, JPQL

Frontend & 3D: React, Next.js, Redux, Zustand, Vite, Tailwind, Streamlit, Three.js, React Three Fiber, OpenGL, Unity, Blender

AI/ML: NumPy, Pandas, scikit-learn, XGBoost, PyTorch, OpenCV, OCR, STT, LLM APIs, Prompt Engineering, Multimodal Pipelines

Testing & DevOps: JUnit 5, MockMvc, PyTest, Locust, Integration & Load Testing, Git, CI/CD, Docker, Linux

Systems & Embedded: Distributed Systems, RAFT Consensus, Event-Driven Architecture, Secure Multi-Party Computation (MP-SPDZ), ESP32/Arduino, ESP-NOW, UDP, Quaternions & Inverse Kinematics